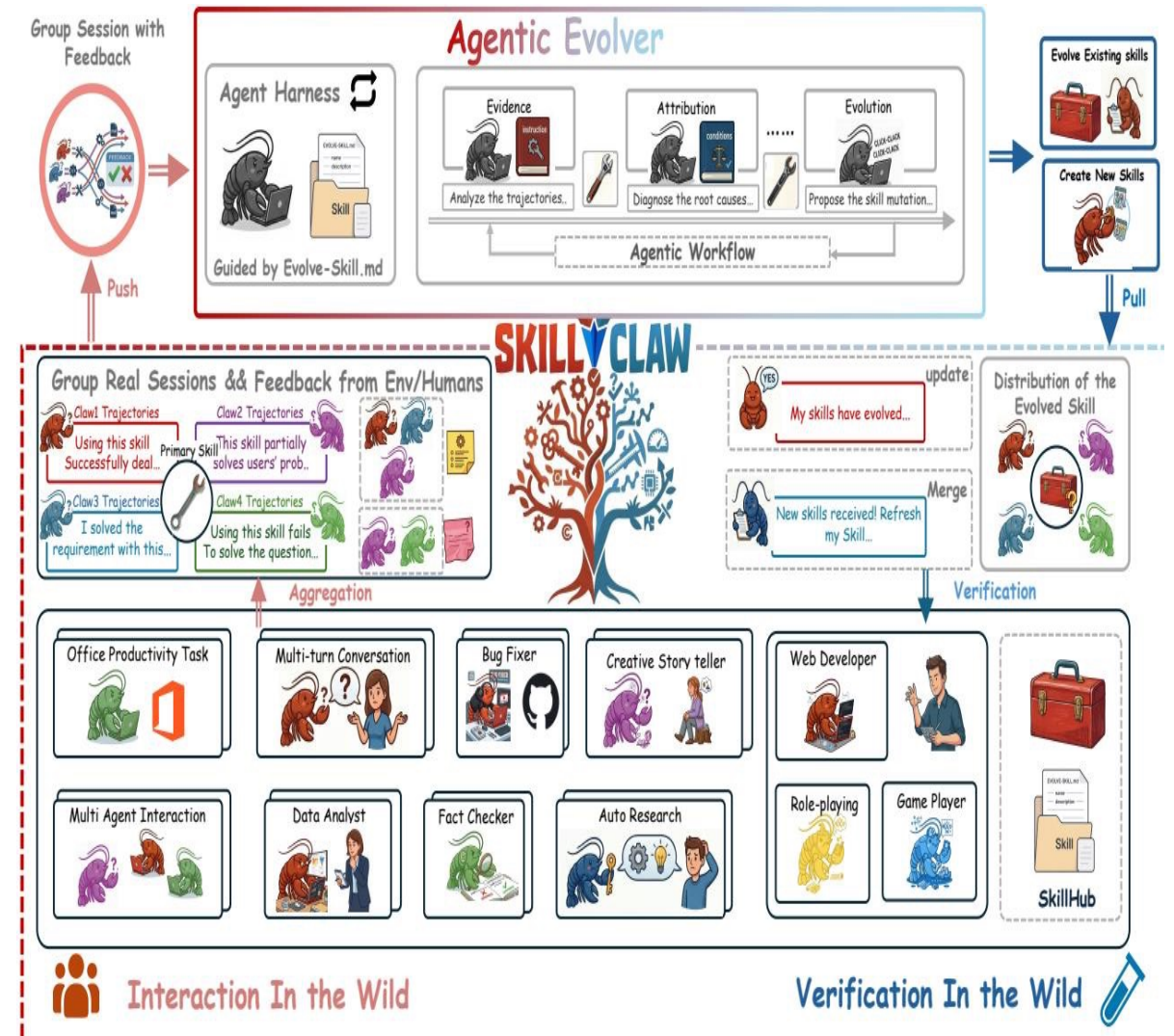


SkillClaw

Let Skills Evolve Collectively with Agentic Evolver

Collective skill evolution for multi-user LLM agent ecosystems — aggregating cross-user trajectories to drive evidence-based, automatic skill improvement.

arxiv.org/abs/2604.08377



Agent Skills Remain Static After Deployment

Users Rediscover Solutions Independently

Static Paradigm (Current)

- Skills manually installed from a centralized hub
- Solutions discovered in-session rarely persist
- Same failures and recoveries repeated across users
- No mechanism for system-level knowledge accumulation
- Each user forced to rediscover solutions independently

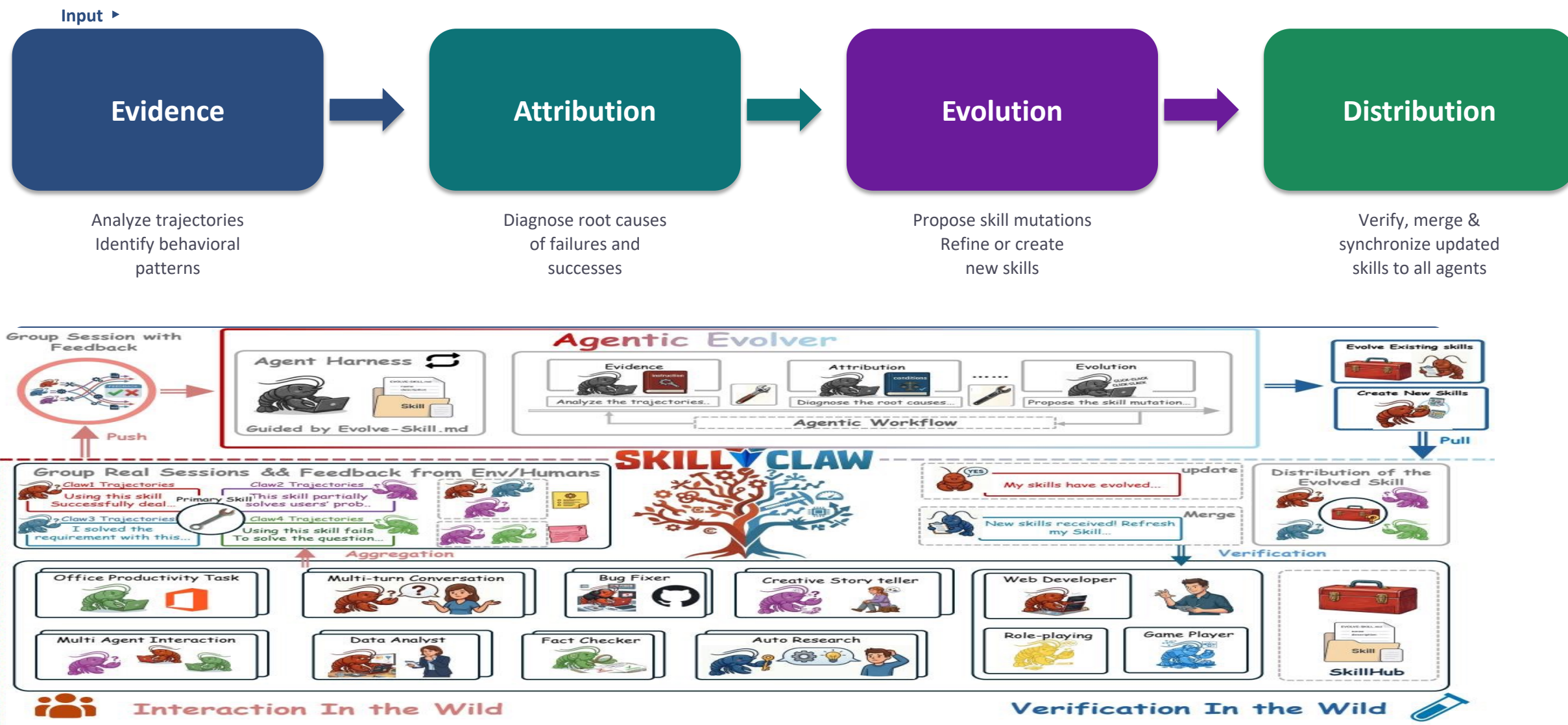


Evolving Paradigm (SkillClaw)

- Skills continuously updated from runtime experience
- Cross-user trajectories feed a shared evidence base
- Agentic evolver proposes verified skill mutations
- Knowledge accumulates at the ecosystem level
- All users benefit from collective improvements

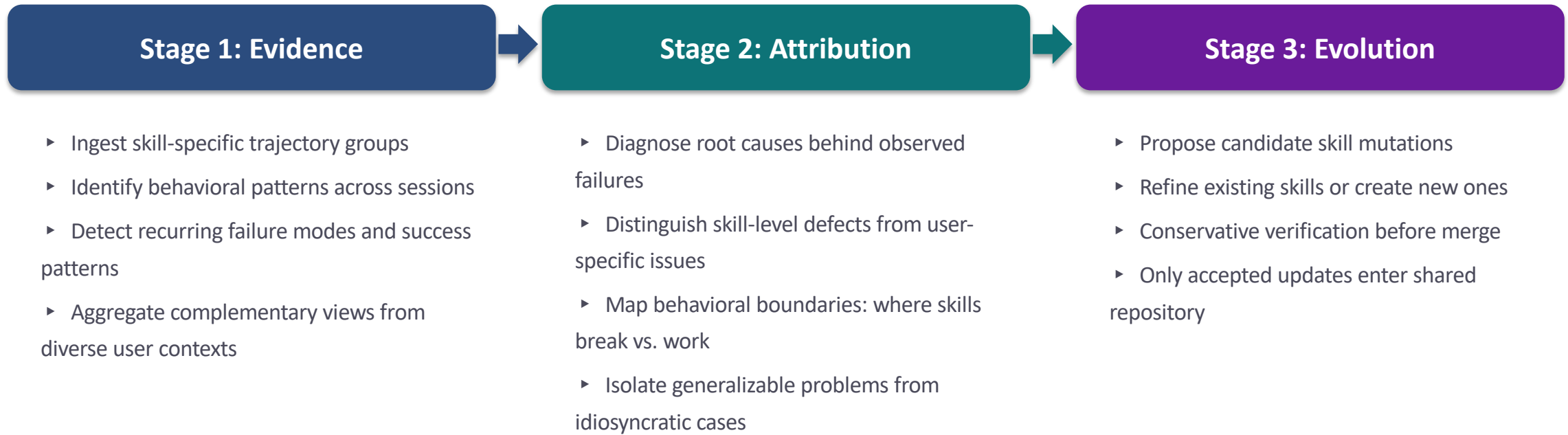
Memory-based methods store trajectories but can't generalize · Skill-based methods compress experience but treat the library as static

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

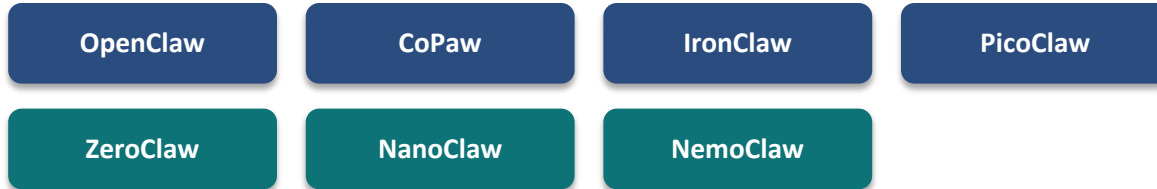
Open-ended agent reasoning over interaction evidence — not predefined update rules



Key distinction: The evolver is an autonomous agent performing open-ended reasoning over evidence — it is not constrained by predefined update rules or fixed mutation operators.

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Compatible Claw-Style Systems



How Aggregation Works

- ▶ Different users exercising the same skill under diverse contexts produce complementary signals
- ▶ Success patterns reveal where a skill works correctly
- ▶ Failure modes reveal conditions under which a skill breaks
- ▶ Together they map the full behavioral boundary of a skill

Single User: Insufficient Signal

A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix. Individual trajectories are sparse and context-specific.

Cross-User Aggregation: Rich Signal

Aggregating trajectories across users exercising the same skill in diverse contexts reveals both conditions under which it works and those under which it breaks — complementary behavioral views.

Trajectories preserve full action–feedback causal chains and are grouped by referenced skills

Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution Attempts

Conservative verification: only 1 of 6 candidates accepted — regressions prevented

Day	Candidate Skill	Skill Function	Validator	Next-Day Action
1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & environment setup	Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Multimodal input validation & output env init	Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Multimodal creative pipeline (extract from PDF/video/image)	Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Added image classification, visual generation, structured validation	Reject	Keep current best pool
5	multimodal-creative-task-pipeline (impr.) + validate-required-input-files	Creative pipeline + per-file fail-fast validation	Reject	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Extended PDF-to-poster / document-to-visual paths	Reject	Not admitted to next cycle

Result: 1 / 6 accepted — The validator rejected candidates that failed to improve over the existing best pool, preventing regressions in the shared skill ecosystem.

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

Successful iterative refinement: 3 of 6 days yielded accepted updates before plateau

Day	Candidate Skill	Change Summary	Validator	Next-Day Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	Accept	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell pitfalls	Accept	Keep current best pool
4	(none)	Same-pool retest; validator confirmed current pool as best	Accept	Continue same best pool
5	git-push-with-auth-fallback	Added "push hang treated as auth failure"; no improvement	Reject	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no improvement	Reject	Not admitted to next cycle

Result: 3 / 6 accepted — Iterative refinement succeeded when evidence supported it. Days 1–3 added meaningful improvements; Days 5–6 proposed changes that didn't improve — correctly rejected.

Key Takeaways: Three Properties That Distinguish SkillClaw

1 Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem. Cross-user trajectory aggregation provides complementary behavioral signals that a single user cannot generate alone.

2 Fully Automatic

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention. The system autonomously collects evidence, attributes causes, and proposes verified skill mutations.

3 Agentic Evolution Paradigm

Skill updates are produced through open-ended reasoning rather than predefined update rules. An autonomous agent analyzes evidence and generates context-aware mutations — not fixed mutation operators.

Implications for Continuously Improving Agent Systems

- Static skill ecosystems are a bottleneck — agents must evolve from their own runtime experience
- Cross-user aggregation transforms isolated sessions into a collective improvement signal
- Conservative verification (1/6 and 3/6 acceptance rates) prevents regressions while allowing genuine progress
- Compatible with multiple Claw-style systems and evaluated on WildClawBench with Qwen3-Max backbone